

MEMORY & FORGETTING

Definition, Types, Models, Levels of Processing,
Interference, Repression and Amnesia

Based on Morgan & King and Standard Academic Psychology Texts

✓ Memory: Definition & Stages	✓ Types of Memory	✓ Models of Memory	✓ Levels of Processing	✓ Forgetting: Interference, Repression & Amnesia
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PART I — MEMORY

Definition of Memory

Memory refers to the mental process through which information is encoded, stored, and retrieved.

According to Morgan and King:

Memory is the process by which information is encoded, stored, and later retrieved when needed.

Memory enables individuals to learn from experience, retain knowledge, recognize people, solve problems, and maintain a sense of personal identity.

Stages of Memory

Memory involves three basic processes:

Stage	Definition	Examples	Notes
1. Encoding	Transformation of sensory information into a form that can be processed and stored.	Repeating a phone number mentally; Understanding the meaning of a sentence.	Types: Visual, Acoustic, Semantic encoding. Semantic encoding is generally the most effective.
2. Storage	Retaining information over time.	—	Involves maintaining information in different memory systems.
3. Retrieval	Bringing stored information back into consciousness.	Recalling examination answers; Remembering a friend's birthday.	Types: Recall (without cues); Recognition (identifying previously learned information).

Types of Memory

1. Sensory Memory

The initial stage of memory.

Characteristics: Very brief duration; Large capacity; Holds raw sensory impressions.

Type	Full Name	Duration	Example
Iconic Memory	Visual sensory memory.	Approximately 0.25–0.5 seconds	Seeing an image briefly after it disappears.
Echoic Memory	Auditory sensory memory.	Approximately 2–4 seconds	Remembering the last words someone said.

2. Short-Term Memory (STM) and 3. Long-Term Memory (LTM)

Short-Term Memory (STM)	Long-Term Memory (LTM)
Temporarily stores information. Capacity: 7 ± 2 items (George A. Miller) Duration: Approximately 15–30 seconds Example: Remembering a verification code.	Stores information for extended periods. Capacity: Potentially unlimited Retention: Long-lasting Includes Episodic, Semantic, and Procedural Memory (see below).

Episodic, Semantic and Procedural Memory

These are subdivisions of long-term memory.

Type	Definition	Characteristics	Examples
A. Episodic Memory Proposed by Endel Tulving	Memory for personal experiences.	Autobiographical; Time and place specific; Involves conscious recollection.	First day at college; Last birthday celebration.
B. Semantic Memory	Memory for facts, concepts, and general knowledge.	Independent of personal experience; Organized knowledge system.	Capital of India; Meaning of psychological terms.

C. Procedural Memory	Memory for skills and actions.	Usually unconscious; Acquired through practice.	Riding a bicycle; Driving a car; Typing on a keyboard.
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Models of Memory

1

Multi-Store Model

Richard Atkinson and Richard Shiffrin



Information moves through rehearsal.

Maintenance Rehearsal	Elaborative Rehearsal
Simple repetition. Example: Repeating a number.	Connecting new information with existing knowledge. Produces stronger long-term retention.

Limitations

- Oversimplifies memory.
- Underestimates active processing.

2

Working Memory Model

Alan Baddeley and Graham Hitch

Component	Function
Central Executive	Controls attention.
Phonological Loop	Processes verbal information.
Visuospatial Sketchpad	Processes visual and spatial information.
Episodic Buffer	Integrates information from different sources.

Levels of Processing Theory

Proposed by: Fergus Craik and Robert Lockhart

This theory argues that memory retention depends on the depth of processing, not merely on rehearsal.

Level	Processing Type	Focus	Example
1	Structural Processing Shallowest level	Focuses on physical appearance.	Is the word written in capital letters?
2	Phonemic Processing Intermediate level	Focuses on sound.	Does the word rhyme with another word?
3	Semantic Processing Deepest level	Focuses on meaning.	What does the word mean?

Conclusion: Deeper processing leads to better retention.

PART II — FORGETTING

Definition of Forgetting

Forgetting refers to the inability to retrieve previously learned information. It may occur due to decay, interference, repression, or neurological damage.

1 Interference Theory

Forgetting occurs because memories compete with one another.

A. Proactive Interference

Definition: Old learning interferes with new learning.

Example: Using an old password instead of a recently changed password.

B. Retroactive Interference

Definition: New learning interferes with old memories.

Example: Difficulty recalling a previous phone number after obtaining a new one.

2

Repression

Sigmund Freud

Painful or anxiety-provoking memories are unconsciously pushed out of awareness.

Characteristics:

- Unconscious process
- Protects the individual from psychological distress
- Commonly discussed in psychoanalytic theory

Example: A person may have difficulty recalling a traumatic childhood event.

3

Amnesia

Amnesia is a pathological loss of memory caused by brain injury, illness, psychological trauma, or neurological disorders.

Type	Definition	Example
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A. Retrograde Amnesia	Loss of memories formed before the injury.	Not remembering events that occurred before an accident.
B. Anterograde Amnesia	Inability to form new long-term memories after brain damage.	Remembering childhood events but forgetting recent conversations.
C. Childhood Amnesia	Difficulty remembering experiences from the first few years of life.	—

Summary

Memory is a dynamic process involving encoding, storage, and retrieval. It includes sensory, short-term, and long-term systems, with long-term memory comprising episodic, semantic, and procedural memories. Major memory theories include the Multi-Store Model, Working Memory Model, and Levels of Processing Theory. Forgetting occurs through mechanisms such as interference, repression, and amnesia, each explaining different aspects of memory failure.